

Supplementary Material

Task-Oriented Semantic Granularity Selection for Bandwidth-Constrained V2V Cooperative Perception

TABLE I

Ego-side online task cues used by the CA-TOSG selector ($d = 21$). All cues are computed locally on the ego vehicle before message selection, removing the causality issue that would arise from depending on the collaborator’s state.

Category	Cue	Definition
Agent context	num_cavs	number of CAVs in scene
	ego_num_objects	ego-detected objects
Frame shape	ego_origin_lidar_shape_0/1	raw LiDAR tensor shape
Range stats	pcd_num_points	total point count
	pcd_mean_range	mean LiDAR range
	pcd_max_range	max LiDAR range
	pcd_std_range	range std. dev.
Distance bins	pcd_near_20m	points within 20 m
	pcd_mid_20_50m	points 20–50 m
	pcd_far_50_80m	points 50–80 m
	pcd_very_far_80m	points beyond 80 m
Quadrant cnt.	pcd_front_points	front-half points
	pcd_back_points	rear-half points
	pcd_left_points	left-half points
	pcd_right_points	right-half points
Long-range	pcd_front_far_30m	front points > 30 m
	pcd_front_far_50m	front points > 50 m
Density bins	pcd_density_0_20	density 0–20 m
	pcd_density_20_50	density 20–50 m
	pcd_density_50_80	density 50–80 m

This document carries material moved out of the main paper for length. It is not a second set of results: the two appendices below are the prior-protocol arms labelled as such in the main text, and the cue table is the definition list of the 21 committed ego-side cues. Section and label names are preserved so that the main text’s pointers resolve against this document.

I. Ego-Side Online Task Cues: full definition list

The selector’s task-cue vector $z_t \in \mathbb{R}^{21}$ is defined in full below; the main paper lists five representative cues and points here.

II. When Are Perception Cues Necessary? LDPC Cliff versus JSCC Graceful Degradation

Protocol note. A prior-protocol arm, not re-evaluated under the frozen P2 protocol: its selector is trained and evaluated inside the codec-comparison pipeline, sharing the retired engine’s BLER lookup and 200-realisation CSI convention. Its numbers may not appear in the same sentence as a frozen-replay number, and no main-text claim rests on them.

Under LDPC + QAM a one-line threshold tracks the learned selector (Section ??); does that equivalence survive a feature codec without a sharp SNR cliff? We replace the feature branch with importance-map JSCC [1] and re-run the per-frame pipeline: per-frame JSCC F1 is re-cached at each SNR (1,000 frames per SNR, AWGN and Rayleigh), the $\{L, \text{Feature-JSCC}\}$ oracle relabelled and the selector retrained—no perception-backbone change.

The mechanism is a codec property. The LDPC + QAM feature F1 cliffs over the 8–12 dB decoding transition (onset 8.0 dB), so SNR is highly informative; the learned JSCC feature F1 is essentially flat at ≈ 0.89 across 0–20 dB, so the estimated SNR says almost nothing about whether a feature message will help on a given frame.

This flips the comparison (Table II). Both codecs are evaluated on the same 2,170 test frames under the identical 200-seed protocol, separating (a) whether the graceful-channel gain exists and resides in the cues—estimated leakage-free in-distribution by 5-fold cross-validation, every frame scored by a selector that never saw it and τ tuned on the training folds only—from (b) whether a deployed, validate-frozen selector realises it, under the headline cross-split protocol.

(a) In-distribution. Under LDPC + QAM the AWGN edge over the best SNR threshold is only +0.002 F1 (95% CI [+0.001, +0.003]): small, and—as the feature ablation shows—not attributable to the perception cues alone (Section ??).¹ Under JSCC the best SNR threshold collapses to the Fixed- L operating point—it cannot gate on an uninformative SNR—and the cue-based edge, measured by the in-distribution k -fold estimator, widens to +0.022 F1 ([+0.020, +0.025]) under AWGN, +0.018 ([+0.016, +0.020]) under Rayleigh, and +0.020 ([+0.018, +0.022]) under OFDM on the test split, and to +0.012–+0.015 on the Culver-City domain shift (all CIs exclude zero). In the separate 200-realisation held-out comparison on validate frames, the cue-based selector recovers 56–62% of the clairvoyant oracle’s headroom over

¹This in-distribution per-frame edge (+0.002) is distinct from two other LDPC-regime quantities: the feature-ablation gain of the cues (Full vs. channel-state-only, +0.0090, 95% CI [+0.0089, +0.0091], Section ??) and the matched-payload margin over the retuned threshold (+0.0032, Section ??). The three measure the selector along the cue, the equal-bandwidth, and the per-frame axis respectively. Under the corrected single-collaborator convention all three are positive, with the ordering $+0.002 < +0.0032 < +0.0090$: the cue axis, smallest and indistinguishable from zero under the retired full-collaborator accounting (−0.0002), is now the largest of the three — weakening the single message widened the gap the cues have to close, and they close part of it.

TABLE II

Learned-selector edge over the best SNR-threshold rule on the same 2,170 test frames (200-seed protocol), reported two ways. In-dist. edge: leakage-free 5-fold cross-validation on the evaluation split (each frame scored by a selector that never saw it, τ tuned on the training folds only). Deployed edge: selector trained on validate and frozen, applied cross-split (the headline protocol). In-distribution the JSCC edge is 9–11 $\times$ the LDPC-cliff edge—an SNR threshold cannot reach it—while under LDPC the margin is the selector’s function form, not a cue gain (cues add no significant F1, Section ??). The deployed JSCC edge is negative: the validate-trained selector does not transfer the graceful-channel gain across the domain shift (Appendix II(b)).

Feature codec	Channel	In-dist. edge	Deployed edge
LDPC + QAM (cliff)	AWGN	+0.002 [+0.001, +0.003]	−0.002 [−0.003, −0.001]
Importance-map JSCC	AWGN	+0.022 [+0.020, +0.025]	−0.004 [−0.007, −0.001]
Importance-map JSCC	Rayleigh	+0.018 [+0.016, +0.020]	−0.008 [−0.011, −0.006]
Importance-map JSCC	OFDM	+0.020 [+0.018, +0.022]	−0.005 [−0.008, −0.003]

Fixed L across the three channels. The two quantities are distinct and are reported separately: under AWGN the oracle headroom is 0.0291 F1 and the selector recovers +0.0181 of it, with 0.0275 and +0.0158 under Rayleigh and 0.0281 and +0.0158 under OFDM. In-distribution, then, the graceful-channel decision is content-bound and carried by the ego-side cues—a gain no SNR threshold reaches on these frames, and one that (b) below shows does not survive the cross-split move.

(b) Cross-split deployment. The signal does not transfer: deployed on test the frozen selector over-selects the feature action (JSCC C-request rate 0.14 $\rightarrow$ 0.42) and falls just below Fixed- L —edge −0.004 (AWGN), −0.008 (Rayleigh), −0.005 (OFDM) on test, ≈ 0 on Culver-City. The gain is present but not portable; capturing it under cross-domain deployment, e.g. by online or self-supervised adaptation [2], is future work.

Whether task-oriented cues are needed is determined by the feature codec’s channel response. Under a cliff codec the decision is channel-bound and SNR suffices; under a graceful codec it becomes content-bound and the ego-side cues—which predict whether object-level detection is already adequate—become the operative signal, in-distribution. Closing the cross-split transfer gap (b) is the remaining challenge.

III. Boundary of the Method: a Second Detection Backbone

Convention. This arm was trained and evaluated on the full-collaborator caches, i.e. before the corrigendum of Change-log P0: it is internally consistent (its own budget, its own N_{cw} , its own references) and its relative conclusion stands, but its absolute F1 and payload figures are on a different accounting basis from the single-collaborator main experiment and are not tabulated beside it.

Positioning. This appendix reports a limitation, not a generality result. It is placed here because its conclusion is negative: the selection procedure developed in the main text is in-sample effective on a second backbone but does not transfer to held-out splits under the equal-budget protocol. No claim in the main text depends on it.

TABLE III

Second backbone (SECOND), equal-budget controlled. Paired $\Delta F1$ of the frozen selector against the SNR-threshold rule tuned to the same budget, over the same 200 CSI samplings; 95% frame-level paired bootstrap intervals. Positive favours CA-TOSG. Second-backbone arm, not deployed; descriptive, no decision is taken from this table.

Split	$B_{\max} = 0.10$	$B_{\max} = 0.20$	$B_{\max} = 0.30$
Validate (in-sample)	+0.0058 [+0.0057, +0.0059]	+0.0095 [+0.0094, +0.0096]	+0.0081 [+0.0081, +0.0082]
OPV2V test	−0.0014 [−0.0015, −0.0013]	−0.0047 [−0.0048, −0.0046]	−0.0066 [−0.0067, −0.0065]
Culver-City	−0.0029 [−0.0030, −0.0028]	−0.0117 [−0.0119, −0.0115]	−0.0167 [−0.0169, −0.0165]

Every result above uses one perception backbone. To test whether the selection procedure—not the particular features—is what carries the behaviour, we repeat the whole pipeline on a second detector, SECOND [3], under an equal-budget controlled comparison: the feature message is charged the same per-frame channel budget as the mainline ($B_F \equiv 0.99$ Msym, the same N_{cw} and the same BLER tables), so the two backbones are compared at equal channel cost. We do not claim that SECOND’s feature tensor compresses to that size, and we declare no codec for it.²

Both SECOND checkpoints are the official OpenCOOD ones, and we verify them before use rather than trusting the file: after a lossless axis-order conversion required by the spconv version in our environment, official inference reproduces the published AP@0.7 to +0.0002 on both splits for the late-fusion model (0.7752 vs. 0.775; 0.6822 vs. 0.682) and to within +0.0019 for the intermediate model. The three per-frame branches (E , L , F) are then re-derived with the same scorer, the same canonical union ground truth and the same IoU-0.5 convention as the mainline, and the entire downstream pipeline—grid expansion, scene-level 9-fold LOSO, the budget walk, and the 200-realisation paired replay—is re-run with the mainline code, each stage first verified to reproduce its own committed mainline product bit-for-bit.

The result is that CA-TOSG is in-sample effective but does not transfer under the equal-budget protocol. Table III gives the paired comparison against an SNR-threshold rule tuned on validate for the same target budget. On validate—the split the selector is trained on—it is ahead at every budget (+0.0058 to +0.0095 F1). On both held-out splits it is behind at every budget, with all confidence intervals excluding zero.

Two mechanisms account for the gap, and both are visible in the arm’s own numbers. First, the L – F margin is narrower on this backbone: SECOND’s object-level branch scores 0.869 / 0.879 / 0.879 mean per-frame F1 on validate / test / Culver-City against a compressed-feature branch

²The transmitted tensor was measured, not assumed: SECOND’s cross-link BEV tensor is 6,758,400 elements per collaborator before compression and 352,000 at the auto-encoder bottleneck. Both are recorded as measurements; neither is used as the operative payload under the equal-budget convention.

at 0.897 / 0.904 / 0.939, so there is simply less to gain from choosing correctly, and the same selection error costs proportionally more. Second, the budget walk selects a different candidate family: all three frozen SECOND selectors carry `class_weight=balanced`, whereas all three mainline selectors carry `class_weight=None`. The reweighted family fits the in-sample class mix well—agreement with the oracle is 0.833 on validate—but transfers poorly, falling to 0.578 on test and 0.534 on Culver-City, where the selector degenerates to near-always- L (E recall 0.027 on test and 0.000 on Culver-City). The E -collapse of Section ?? is therefore worse on this backbone, not milder: under Rayleigh the selector’s ρ_E is 0.002 (test) and 0.000 (Culver-City) while the oracle spends 0.157 and 0.133. The large payload reductions this arm shows (67–90%) follow from that collapse to L and are not evidence of good selection.

We report this as a boundary of the present method rather than a tuning target: nothing was retrained and no threshold was adjusted after seeing these numbers.

IV. Material moved from the main paper

The four subsections below are reproduced verbatim from the main paper’s results section; the main text keeps a short summary of each and points here.

A. Collaboration Is Not Always Beneficial

Collaboration is not unconditionally beneficial, and that is what makes an explicit per-frame selector necessary rather than a convenience. Two failure modes call for two responses. When the channel cannot carry a feature message, requesting one spends the collaborator’s transmission budget for nothing and collapses the output to the ego-only floor (the ego vehicle’s own pre-fusion detection); the design answer is already in place – the oracle removes any action whose frame-level failure probability is ≥ 0.999 from its feasible set (the same 0.999 constant as the §?? mask), and on failure the pipeline reverts to the ego-only output rather than a phantom feature. When the channel can carry the message, requesting it can still cost accuracy: on the easy stratum the selector’s realised output falls below even the always-object-level (Fixed- L) baseline.³ This mode has no masking answer; its remedy is left to future work. Two CSV-verified quantifiers bound where the ego-side harm sits: the ego-only output strictly exceeds the object-level fused output on 1.5 / 5.8 / 0.2% of validation / test / Culver-City frames, and – from the same per-frame (comp – ego) identity, re-derived on

³Test Easy stratum (top tercile of the ego’s own object-level F1, the §?? stratification), evaluated under a deterministic reliable-channel condition (AWGN 16 dB; frame-level BLER ≈ 0 , well above the 8.0 dB onset), isolating the difficulty axis from channel variability: the frozen selector’s realised F1 at $B_{\max} = 0.20$ is 0.9749 vs the Fixed- L baseline 0.9796 – a gain of -0.0047 (frame-level paired 95% CI $[-0.0074, -0.0024]$; $n = 713$ frames). The selector requests F on 241 of these 713 frames; on the remaining 472, where it requests L , its output is frame-identical to Fixed- L , so the paired difference arises entirely on the F -request frames – the loss is a structural consequence of requesting features on already-easy frames.

the corrected single-collaborator caches – the compressed-feature message, when delivered, yields lower frame F1 than the ego-only fallback on 1.3 / 6.5 / 0.9% of frames. Test carries the harm most, consistent with fusion having the least to add in thin scenes (mean 15.2 ground-truth objects on test vs 27.8 on validate and 41.0 on Culver-City). No new signalling is needed to act on this: E (ego-only, issue no request) is already a deployed action of $\mathcal{S}$ with its own codepoint, so the remedy is a selection-policy question—the frozen selectors almost never choose it (Section ??)—not a message-format one.

B. Comparison with Where2comm

To place CA-TOSG beside the most prominent spatial-selection cooperative perception method, we report a reference-only comparison against Where2comm [4] on the same OPV2V validate split (1,980 frames). Three caveats bound what it can support, and they are stated rather than buried: the numbers come from our own 50-epoch reproduction, not the authors’ released model; they are measured under a perfect channel, so no part of this paper’s transport model applies to them; and they were scored by the retired global-sort scorer, which is a different track from the frozen AP chain that produces every headline number here. Our reproduction reaches $\text{AP@0.5} = 0.871$ under a perfect channel, and we deliberately draw no ranking from it. It is not comparable to any number in this paper, on three independent axes at once: a different scorer (the retired global-sort track, not the frozen AP chain), a perfect channel rather than the modelled transport, and—decisively—a different collaborator convention, since the reproduction fuses every collaborator while the corrected main experiment pays for one. The direction of such a comparison is not even stable: under the corrected accounting the validate Fixed- L reference is 0.7819, below 0.871, which reverses the ordering an earlier version of this section reported. Numbers measured under three different conventions cannot be ordered, and we do not order them.

The comparison is also orthogonal to CA-TOSG: Where2comm is a spatial-mask method that decides which feature regions to send within a fixed feature-level granularity, whereas CA-TOSG decides whether to send feature-level information at all and at what granularity, given the channel. The two compose: Where2comm can be deployed inside the C_q branches of CA-TOSG as an alternative feature codec, while the granularity selector decides when feature-level communication is worth its payload and channel risk. Under a bandwidth-constrained, time-varying link, even a feature-level message is realised only when the channel can deliver it—precisely the regime CA-TOSG arbitrates.

C. Collaborator Scale

The evaluation so far treats the collaborator set as given. To ask how the policy behaves as more vehicles are reachable, we sweep the number of requested collaborators

$N \in \{1, 2, 3\}$, keeping the frozen selector, λ^* and τ^* untouched.

N is not the operative quantity: what matters is the effective count k_{eff} , since requesting three cannot conjure a third. Of the 1,980 validate frames, 733 are saturated at $N = 1$, 1,081 at $N = 2$ and 1,234 at $N = 3$; on Culver-City the supply runs out, so $N = 2$ and $N = 3$ are the same arm. Comparisons are conditioned on the same frames.

F1 then rises with k_{eff} with sharply diminishing returns. At $B_{\text{max}} = 0.20$ on validate, $N = 1 \rightarrow 2$ buys +0.0300 F1 for +0.1183 Msym and the third adds +0.0061 for +0.0991 Msym—a fifth of the gain for three-quarters of the channel; on test the pattern is flatter (+0.0096 then +0.0008). The payload sub-linearity is a supply effect: the action distribution is unchanged across N , so the extra channel use comes only from frames that gained a collaborator.

Because “request N ” is ambiguous when fewer than N collaborators exist, we bracket the two delivery semantics—charging only for what is delivered, versus charging for what was requested—on validate at $N = 2$, on the 964 frames where the two differ. The bracket is tight: charging for what is delivered rather than for what was requested raises realised F1 by +0.00016, +0.00036 and +0.00099 across the three budgets, at identical payload—the request is what is charged under both—so the conclusions here do not depend on which convention is adopted. Finally, the budget does not transfer automatically. The selector is frozen against an average per-frame budget measured with a single collaborator; total communication grows with the number of collaborators actually served, so the original budget guarantee does not carry over unchanged. This is measurable rather than hypothetical: on validate at $B_{\text{max}} = 0.20$, $N = 3$ spends 0.36842 Msym per frame, well above the 0.20 it was frozen to satisfy — and under the corrected single-collaborator accounting even $N = 2$ (at 0.26937) already breaches it. A deployment that serves more collaborators must therefore re-freeze, or budget on k_{eff} rather than per collaborator.

D. Deployment Robustness and Cost

Three practical imperfections—a noisy SNR estimate, channel-state aging, and a stale request—degrade the selector only gracefully, because an uncertain channel state falls back to L (Table IV): SNR-estimation noise up to $\sigma \approx 1$ dB costs ≤ 0.0002 F1; a Jakes model at 60 km/h decorrelates the estimate within ~ 10 ms, costing -0.0025 before plateauing; a one-frame-stale decision costs at most -0.0109 under our i.i.d.-per-frame assumption (changing 18.2% of decisions). The near-immunity to estimation noise follows from the channel-type-anchored policy of Section ??: perturbing continuous $\hat{\gamma}_t$ rarely flips a decision anchored on the discrete channel-type flag, and under Rayleigh the action remains L regardless. Finally, the deployed Random Forest runs in 52.1 ± 5.6 ms per frame (P95 = 58.3 ms) on a single CPU core, i.e. the selector’s own inference occupies about half of the 100 ms

TABLE IV

Deployment robustness (OPV2V test; realised frame F1). Each row perturbs one imperfection while holding the others ideal; losses are bounded because an uncertain channel state defaults to the robust L message. Re-derived under the single-collaborator convention with the frozen selectors at $B_{\text{max}} = 0.20$ (results/sensitivity/robustness_frozen.csv); the CSI-noise and Jakes models are the ones used previously, so only the selector, grid and utilities changed.

Imperfection	Operating point	$\Delta F1$
SNR-estimation noise σ	≤ 1 dB	-0.0002
	2 dB	-0.0006
	5 dB	-0.0021
CSI aging (Jakes, 60 km/h)	stale by ~ 10 ms	-0.0025 (then plateaus)
Request delay	1-frame-stale decision	-0.0109 (pessimistic i.i.d. b.

frame interval of a 10 Hz LiDAR cycle; we measure the selector only and make no claim about the end-to-end perception–transmission chain. The figure is the slowest of the three frozen selectors over 1,000 batch-1 trials each; the deployment claim must hold for every budget, so the worst case is the one quoted.

A fourth imperfection concerns the cheap message rather than the expensive one. The utilities above model the object-level link as reliable ($\text{BLER}_L = 0$), which is an assumption about the low-rate path, not a measurement. Re-weighting the object-level utility analytically as $\text{eff}'_L = \text{late}(1 - \text{BLER}_L) + \text{ego BLER}_L$ over $\text{BLER}_L \in \{0.01, 0.05, 0.10\}$, with every policy frozen and blind to it, leaves the payload — and therefore the payload advantage — untouched by construction, and costs F1 roughly linearly: on test at $B_{\text{max}} = 0.20$, $\text{BLER}_L = 0.10$ costs the selector 0.00634 F1, the threshold rule 0.00588 and Fixed L 0.00734 (results/sensitivity/r23_object_message_bler.csv). The selector’s margin over Fixed L therefore widens as the cheap link degrades, while its gap to the threshold rule widens against it, from -0.00010 to -0.00056 — the threshold rule leans harder on the feature action ($\rho_F = 0.20$ against 0.12) and is correspondingly less exposed to an unreliable L . We report this rather than the favourable half alone; it remains inside the pre-registered margin of 0.005 at every BLER_L evaluated.

References

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